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Planting time and variety effects on biomass, harvest index, and yield of irrigated soybean in mid-Southern United States

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Abstract

Soybean [*Glycine max* (L.) Merr.] biomass and grain yield has increased over the past several decades in the mid-southern United States. This is attributable to technological advances and improved management strategies. However, a better understanding of biomass accumulation and partitioning is needed to improve our knowledge base of varietal growth habits relative to yield, planting date, and harvest index (HI). Field experiments within a split plot arrangement in a randomized complete block design were established in 2017 and 2018 in Stoneville, MS. The study aimed to evaluate the effect of early (late-April or mid-May) and late (late-May) planting on biomass, HI, and yield amongst eight soybean varieties. Soybean total biomass accumulation was collected at multiple development stages, including V4, R2, mid R5, mid R6, and R8, and partitioned into senesced leaves, pods, and seeds. Overall, the planting date had no effect on yield, HI, and biomass accumulation at any of the growth stages. Yet, the interaction between planting date and variety significantly affected biomass accumulation at the mid R5 stage. Contrarily, the variety selection significantly affected yield, HI, and biomass accumulation at all growth stages except mid R6. The total biomass accumulation at R8 was greatest for Asgrow 46X6, Asgrow 4632, Terral 4857X, Terral 48A76, and Credenz 4748, when pooled over planting dates. Averaged across two planting dates, the greatest yield was produced by Terral 48A76, Asgrow 4632, and Asgrow 46X6. Furthermore, averaged across site-years, HI was greatest for Asgrow 4632 and Terral 48A76. Based on the results of this study, evaluating soybean HI rather than overall biomass accumulation may be more beneficial for variety selection decisions.

Plain Language Summary

Soybean is an important crop for mid-southern United States. Soybean grain yield has increased over the past several years due to improved management strategies.

Abbreviation: HI, harvest index.

However, our understanding of the varietal differences in terms of physiological parameters on which variety selection decisions should be made are limited. This study compared eight different soybean varieties planted either early or late and found that biomass accumulation and grain yield differed by variety. Our key finding is that harvest index is a much better indicator for variety selection decision rather than biomass accumulation.

1 | INTRODUCTION

Exploring biomass accumulation and partitioning patterns are fundamental for understanding and improving yield potential in soybean [*Glycine max* (L.) Merr.] (Yang et al., 2022). Advancements in genetics have led to the introduction of short-seasoned soybean varieties with enhanced dry matter accumulation and the harvest index (HI) to the extent of 78% and 22%, respectively, aiding in increasing yield (Kumudini et al., 2001). Huang et al. (2009) opined that the yield has direct link with the dry matter accumulation and this correlation strengthens as plant approaches maturity.

Modern soybean varieties produce greater biomass and grain yield than soybean cultivated 50 to 100 years ago (Pedersen & Lauer, 2004). The breeding efforts have expanded the pod-filling duration in modern varieties while keeping the sowing-to-flowering duration unchanged, enhancing both biomass and HI (Yang et al., 2022). However, these breeding advancements have also reallocated the assimilate distribution, channeling them toward the pods/seeds than the stems, affecting HI (Yang et al., 2022). This is partially because the modern variety of soybean needs substantial amount of nutrients to generate higher biomass, and grain. Specifically, the demand of nutrients such as nitrogen, phosphorus, sulfur, and copper is high, resulting in a greater HI (Bender et al., 2015). Also, evidence in literature indicated that HI has a greater influence on soybean yield improvements compared to biomass accumulation (Cui & Yu, 2005). Hence, studying biomass accumulation and HI along with yield can inform trait-targeted breeding tactics, as yield increases by enhancing all the yield aspects synchronously or by substantially improving one or two components (Cui & Yu, 2005).

Research has shown that plant biomass production and HI in soybean are significantly influenced by planting dates and variety selection, amongst other factors (Bender et al., 2015). The early planting date is generally associated with an increased biomass, seed/pod number, HI, and yield while the late planting dates results in a compressed vegetative/reproductive cycle, lower growing degree days or radiation uptake by plants, ultimately reducing yields (Bhatia et al., 1999; Chen & Wiatrak, 2010; Kessler et al., 2020; Sadeghi & Niyaki, 2013). The planting date effects are usually uniform across the locations. However, the varietal effects differ by location exhibiting varying HI, where older cultivars sometimes performs better than the newer ones (Kessler et al., 2020; Pedersen & Lauer, 2004). Therefore, the variety selection is vital, as the response of the varieties differ due to planting dates and climatic conditions (Egli & Bruening, 2000). Literature also indicated that the interplay of variety and planting date alters the yield components including biomass, HI, pods plant^{-1}, seed weight, and quality (Bhatia et al., 1999; Sadeghi & Niyaki, 2013). Moreover, the stoichiometry of dry matter (biomass) and HI changes within the plant as it approaches R5 growth stage. At R5, the dry matter tends to increase while HI decreases because of the slowdown in node production retarding pod/seed generation efficiency (Board & Maricherla, 2008). Therefore, yield advances associated with common agronomic practices, including variety selection and adjusting planting date require a deeper understanding of soybean's aboveground biomass production and HI dynamics (Koester et al., 2014).

The southern region of the United States including Mississippi, adopted the early season production system (ESPS) as a management tool for soybean (Olomitutu et al., 2024). The system increases yield by taking advantage of favorable climatic conditions and avoiding reproductive stage heat and drought stress (Heatherly & Elmore, 2004). The ESPS is a combination of planting in early spring, using early-maturing, indeterminate varieties (Heatherly & Elmore, 2004). Although the previous research has suggested late April to be the optimal planting time for Mississippi soybeans, the planting gets delayed because of wet soils resulting from excess precipitation events and inability to accommodate planting increased soybean acreage within the optimal planting window (Bateman et al., 2020; Chen & McClendon, 1984; MSPB, 2019).

Therefore, testing different varieties at multiple planting windows could provide valuable information to Mississippi soybean producers. Our hypothesis is that soybean varieties will respond differently in terms of biomass accumulation, HI, and yield when planted at different dates. The objective of this research was to assess the varietal difference in biomass accumulation, HI, and soybean yield at two planting dates. Moreover, this study will allow varietal selection decisions based on growth habits, yield, and HI across multiple planting dates.

TABLE 1 Sites for soil test data for April and May planted soybean at the Delta Research and Extension Center in 2017 and 2018.

Site-year	pH	OM	P	K	S	Ca	Mg	Mn	Zn
		ppm							
April 2017	6.97	1.55	47	157	25	1149	211	44	1.25
May 2017	7.08	1.65	41	169	51	1223	232	48	1.28
April 2018	6.72	1.06	41	192	31	1077	202	60	1.25
May 2018	7.48	1.45	25	147	90	3719	541	42	1.21

2 | MATERIALS AND METHODS

2.1 | Description of sites

Field experiments were conducted in 2017 and 2018 to determine the biomass accumulation and partitioning of soybean in the mid-southern United States. Eight modern varieties were tested at two different planting dates. Experiments were established at three sites at the Mississippi State University Delta Research and Extension Center near Stoneville, MS, with a history of high soybean yields (>70 bushels acre^{-1}). Experimental sites represented a silt loam and a clay soil that are commonly cropped to soybean in Mississippi.

In 2017, at the Dubbs silt loam (Fine-silty, mixed, active, thermic Typic Hapludalf; N 33˚26.02′, W 90˚54.34′) site, corn (*Zea mays* L.) was the previous crop and at the Tunica clay (Clayey over loamy, smectic over mixed, superactive, nonacid, thermic Vertic Epiaquept; N 33˚25.05′, W 90˚54.19′) site, soybean was the previous crop. In 2018, corn was the previous crop for the Dubbs silt loam (Fine-silty, mixed, active, thermic Typic Hapludalfs; N 33˚24.54′, W 90˚54.51) site, and rice (*Oryza sativa* L.) was the previous crop for the Tunica clay (Clayey over loamy, smectic over mixed, superactive, nonacid, thermic Vertic Epiaquept; N 33˚25.05′, W 90˚54.19′) site. Eight composite soil samples (two per replicate) were collected from the 0-to-6-inch depth at each experimental site before planting. Each composite sample consisted of eight, 1-inch diameter cores. Soil samples were oven-dried for 48 h, crushed to pass through a 0.07-inch sieve, and extracted with Lancaster extractant (Cox, 2001). Lancaster extracts were analyzed using inductively coupled atomic plasma spectroscopy (ICPAES) (Soltanpour et al., 1996). Soil water pH was determined in a 1:2 soil/water (w/v) ratio using a glass electrode. All soils contained a minimum of 59 lbs P_2O_5 acre^{-1}, 100 lbs K_2O acre^{-1}, and 10 lbs S acre^{-1} ensuring that nutrients were not yield limiting. Soil test data were collected to effectively manage nutrient needs specific to soybean (Table 1). Weeds were controlled with herbicides to maintain a weed-free environment. Same planting and harvesting machinery were consistently used at all locations and years. Plots were monitored using Watermark soil moisture sensors (IRROMETER) and irrigated each year as needed to ensure biomass accumulation was not limited due to water availability.

2.2 | Planting date and varieties

The experimental design for all studies was a split plot with four replications. Whole-plot treatments were planting date (PD) and consisted of early (late-April or mid-May) and late (May), and subplot treatments included eight soybean varieties. Soybean maturity group 4 varieties included (1) ‘Asgrow 46X6’, (2)‘Asgrow 4632’ (Bayer CropScience), (3) ‘Terral 4857X’, (4)‘Terral 48A76’, (5) ‘Hornbeck 4953’, (6) ‘Credenz 4748’ (BASF Plant Science), (7) ‘University of Arkansas 5014’ (University of Arkansas Foundation), and (8) ‘USDA DS 25-1’. These eight soybean varieties were selected as they are frequently cultivated in the study region. Soybean was seeded into conventionally tilled seedbeds at 135,000 seed acre^{-1} utilizing small plot John Deere vacuum planter (Deer and Company). Thinning was performed at all site years 12 to 20 days after planting for ensuring uniformity in stand count and to align with the required plant population. Each plot consisted of eight rows of soybean spaced 40 inches apart separated by a perpendicular 10-ft alley. Individual plots, measuring 26.75-ft wide by 35 ft-long were flagged to establish plot boundaries. Four rows of each plot were designated as areas for destructive sampling and the middle two rows of the remaining rows were harvested for seed yield analysis.

2.3 | Measurements and methods

Aboveground biomass samples were collected at the V4, R2, mid R5, mid R6, and R8 development stages to determine biomass accumulation, partitioning, HI, and yield (Table 2). Vegetative stages were determined by counting the number of nodes on the stem containing a fully developed leaf, beginning with the unifoliate node (Fehr et al., 1971). To assess soybean biomass production, whole aboveground portions of soybean plants were collected from a 3.2-ft section of one row within the four rows dedicated to destructive sampling. For R5 measurements, samples were harvested and pods removed and separately weighed after drying. Seed from R6 and R8

TABLE 2 Selected dates for agronomic management and destructive biomass sampling at the Delta Research and Extension Center in 2017 and 2018.

Site-year	Planting date	V4 Biomass samples	R2 Biomass samples	R5 Biomass samples	R6 Biomass samples	R8 Biomass samples
	Month/day	——days after planting——				
April 2017	5/7	29	45–51	80–87	111–118	132
May 2017	5/17	34	48–55	81–83	102	132
April 2018	4/25	30	48	76–96	104–117	140–145
May 2018	5/16	27	47	78-89	100–111	124–140

samples were separated from pod in addition to pod from stem. Senesced leaves were collected at R5, R6, and R8 to quantify biomass production for senesced leaves. Senesced leaves were collected by attaching 3.2 ft of landscape fabric (Lowe's) to surveyor stakes to form a net to catch the leaves (data not shown). After collection, aboveground portions of soybean plants were oven dried (Forced draft, 149˚F) for a minimum of 48 h and mass for each sample was determined using an analytical balance (Denver Instrument Company). All portions of dried samples were weighed to determine aboveground biomass. The combined weight of all samples was used to determine HI, calculated by dividing seed yield by total aboveground biomass.

Subsamples of each portion of biomass were collected and ground to pass through a 0.07-inch sieve (Thomas Scientific 1654 High Hill Road Swedesboro, NJ 08085). After grinding, samples were digested in HNO_3 and 30% H_2O_2 and analyzed for elemental concentrations by ICP-AES (Jones & Case, 1990). A small plot combine (Kincaid Equipment) was used and two rows were harvested from the four rows designated for harvest from each plot. Soybean seed yields were collected. This collected seed yield was adjusted to standard moisture level (13%) to compute the adjusted yield as per the formula mentioned in Equation 1. This adjusted yield was then used for computing HI and for yield analysis.

$$\text{Adjusted yield} = \text{actual yield} \times \frac{(100 - \text{actual moisture content})}{(100 - \text{standard moisture content})} \quad (1)$$

2.4 | Data analysis

Development stages were designated as days after planting for each site year. Reproductive development stages were divided into plant components for separate analysis containing senesced leaves, pods, seeds, and remaining aboveground biomass. The remaining biomass is referred to as stem plus leaves, which includes stem, petioles, and attached leaves (data not shown).

Data were combined over years 2017–2018 and subjected to analysis of variance (ANOVA) using a general linear mixed effects model using PROC Glimmix procedure in SAS v. 9.4 (SAS Institute). The model is most competent in modeling the nested random effects to represent the split plot design more accurately (Stroup, 2012). The current study's split plot design is inherited with the hierarchal structure containing the nested random effects, for instance, sub-plots are nested within blocks, and blocks are nested within main plots (Schielzeth & Nakagawa, 2013). Better competency in modeling these random effects adds to the more reliable estimation of the study's fixed effects. Also, this study model is considered superior in comparison with the traditional ANOVA, in controlling the type 1 error probability and modeling the intricate variance structure, found in split plot design (Matuschek et al., 2017).

The model followed in the current study is further elaborate in Equation 2:

$$Y_{abc} = \mu + \mathrm{PD}_a + \mathrm{VAR}_b + (\mathrm{PD} \times \mathrm{VAR})_{ab} + \mathrm{Rep(PD)}_{ac} + \epsilon_{abc} \quad (2)$$

Where, PD is planting date and VAR is variety. Y_{abc} denotes the measured dependent variable for the c-th replication, for the a-th level of PD and b-th level of VAR. Moreover, μ is the overall mean value, PD is the whole plot factor, and VAR is the subplot factor. Overall, PD, VAR, and their interactions were considered as fixed affect, and replication nested within the whole plot factor was considered as random variable. Lastly, ϵ_{abc} represents the residual error term.

Biomass accumulation, yield, and HI (lb bu^{-1}) across development stages V4, R2, R5, R6, and R8 were subjected to ANOVA. After performing ANOVA, the least square means were calculated, and mean separation ($p < 0.05$) were based on Fisher's protected Least Significant Difference (LSD) test. The LSD value guiding this test was calculated using Equation 3 (Saxton et al., 1998):

$$LSD = t_{\alpha/2,\mathrm{df}_{\mathrm{residual}}} \sqrt{\frac{2 \cdot MSE}{n}} \quad (3)$$

TABLE 3 Analysis of variance *P*-values for total biomass, harvest index, and yield at the V4, R2, mid R5, mid R6, and R8 development stages.

	Total biomass					Harvest index	Yield
Effect	V4[a]	R2[b]	Mid R5[c]	Mid R6[d]	R8[e]	R8	R8
Planting date	0.9497	0.9431	0.8319	0.0995	0.4329	0.6692	0.2347
Variety	**0.0006**	**0.0003**	**<0.0001**	0.3567	**0.0024**	**<0.0001**	**<0.0001**
Planting date × variety	0.4394	0.5815	**0.0496**	0.4781	0.8836	0.1091	0.1589

Note: Values in bold are significant.
[a]V4 = Total biomass at V4 development stage.
[b]R2 = Total biomass at R2 development stage.
[c]Mid R5 = Total biomass after R5 development stage.
[d]Mid R6 = Total biomass after R6 development stage.
[e]R8 = Total biomass at physiological maturity.

Where, $t_{\alpha/2,\mathrm{df}_{\mathrm{residual}}}$ denotes benchmark value for t-distribution at $\mathrm{df}_{\mathrm{residual}}$ degree of freedom and α level of significance. MSE represents the mean square error previously calculated by ANOVA, and n signifies the number of replications.

The values of the differences between means of all paired groups were compared to the LSD value computed by Equation 3. If the difference between two groups is greater than LSD, then the two means were deemed significantly different, whereas in all other cases, the two means were considered not significantly different.

3 | RESULTS

3.1 | Total aboveground biomass

Variety and planting date interactions were noted at R5 in affecting the biomass accumulation (Table 3). Averaged across planting date, Terral 4857X, Terral 48A76, Asgrow 46X6, and USDA 25-1 produced the greatest total biomass at V4. (Table 4); however, only Terral 48A76 was different from all remaining varieties. At R2, USDA 25-1, U of A 5014, and Terral 4857X produced the greatest total biomass compared to all other varieties (Table 4), with only USDA 25-1 being different from all remaining varieties. The total R5 biomass ranged from 5418–7271 lbs acre^{-1} (Table 5). At this stage, U of A 5014, planted in April and Credenz 4748 and USDA 25-1 planted in both April and May produced total biomass greater than the other varieties (Table 5). Terral 48A76 produced the least biomass at R5 (Table 5). Only the main effect of variety influenced total biomass at V4, R2, and R8 (Table 3).

At R8, Asgrow 46X6, Asgrow 4632, Terral 4857X, Terral 48A76, and Credenz 4748 produced the greatest total soybean biomass when averaged across planting date (Table 4). Hornbeck 4953 produced the least total soybean biomass. Asgrow 46X6 produced total soybean biomass at R8 18% more than Hornbeck 4953 (Table 4).

TABLE 4 The influence of variety (VAR) on total biomass accumulation collected at the V4 through R8 development stages for research conducted at the Delta Research and Extension Center near Stoneville, MS, in 2017 and 2018.

	Biomass accumulation		
VAR	V4 total	R2 total	R8 total
Asgrow 46X6	435 ab	2098 cd	9747 a
Asgrow 4632	413 bc	1950 cd	9398 ab
Terral 4857X	434 ab	2189 abc	9001 ab
Terral 48A76	482 a	1997 dc	9179 ab
Hornbeck 4953	364 cd	1806 d	8041 c
Credenz 4748	357 d	2105 bcd	9176 ab
U of A 5014	371 cd	2444 ab	8964 b
USDA 25-1	448 ab	2507 a	8808 b
P value	0.0006	0.0003	0.0024
LSD	50	288	739

Note: Means within a column followed by the same letter are not different at $P \leq 0.05$.

3.2 | Yield and harvest index

Soybean yield was influenced only by variety (Table 3) and yields ranged from 45.2-63.7 bu acre^{-1}. Averaged across planting date, Terral 48A76 and Asgrow 4632 produced greater grain yields than U of A 5014, Credenz 4748, Terral 4857X, Hornbeck 4953, and USDA 25-1 (Table 6). USDA 25-1 produced the least soybean yield when averaged across planting date (Table 6). Moreover, in this study planting date had no influence on soybean yield.

Only the main effect of variety influenced HI (Table 3). Soybean at harvest produced HI ratios ranging from 30.2% to 38.2% (Table 6). Averaged across planting date, Terral 48A76 and Asgrow 4632 produced HI greater than the other varieties (Table 6). Credenz 4748 and USDA 25-1 produced the least harvest index percentages at 30.2% and 31.1%, respectively (Table 6).

TABLE 5 The influence of variety × planting date interaction on total biomass accumulation at the mid R5 development stage for research conducted at the Delta Research and Extension Center near Stoneville, MS, in 2017 and 2018.

Planting date	Variety	Mid R5 total biomass
		lbs acre^{-1}
1	Asgrow 46X6	5929 e–l
1	Asgrow 4632	5248 m
1	Terral 4857X	5785 g–m
1	Terral 48A76	5174 m
1	Hornbeck 4953	6013 e–j
1	Credenz 4748	6900 abc
1	U of A 5014	7271 a
1	USDA 25-1	6830 a–d
2	Asgrow 46X6	6448 b–f
2	Asgrow 4632	6076 e–h
2	Terral 4857X	5418 g–m
2	Terral 48A76	6010 e–k
2	Hornbeck 4953	6042 e–i
2	Credenz 4748	6574 a–e
2	U of A 5014	6095 d–g
2	USDA 25-1	7105 ab
P value		0.0496
LSD		756

Note: Means within a column followed by the same letter are not significantly different at $P \leq 0.05$.

TABLE 6 The influence of variety on grain yield and harvest index (HI) collected at R8 development stage for research conducted at the Delta Research and Extension Center near Stoneville, MS, in 2017 and 2018.

Variety	Yield	HI
	bu acre^{-1}	%
Asgrow 46X6	60.83 ab	34.1 b
Asgrow 4632	62.30 a	37.2 a
Terral 4857X	56.26 cd	34.1 b
Terral 48A76	63.70 a	38.2 a
Hornbeck 4953	53.33 d	33.8 b
Credenz 4748	57.01 c	30.2 c
U of A 5014	58.06 bc	34.1 b
USDA 25-1	45.26 e	31.3 c
P value	0.001	0.001
LSD	3.49	2.14

Note: Means within a column followed by the same letter are not different at $P \leq 0.05$.

4 | DISCUSSION

The goal of this research was to provide a greater understanding on the dynamics of biomass production, HI, and yield with respect to planting dates, and to provide data for aiding researchers and producers in variety selection decisions for irrigated soybean in the mid-southern United States. Although the eight varieties in this study are frequently grown in this region, variability in the rate of biomass accumulation across plant components was expected due to a range of environmental conditions and germplasms (Bender et al. 2015). Bender et al. (2015) reported an increase in biomass accumulation and yield potential in modern soybeans varieties. Moreover, the greater biomass production is expected to influence yield potential by boosting photosynthesis and thereby contributing to higher seed yields. Previous research by Marcelis (1996), Engels et al. (2012), and Bender et al. (2015) reported greater biomass accumulation throughout the growing season resulting in increased HI and dry weight of plant components influencing nutrient partitioning.

Overall, literature offers mixed opinion on the effect of planting dates and variety on soybean yield and other attributes such as biomass generation and HI. Specifically, varietal differences are noted to be significant in influencing soybean, also supported by the current findings (Ray et al., 2008). Within the scope of this research, differences of biomass accumulation are attributed to genetic isolines of the varieties tested (Cober & Morrison, 2010). Planting date and relative maturity for these varieties ranged from 4.6 to 5.0, likely not affecting variability in biomass accumulation. Differences in relative maturity for the varieties in this study vary from 5 to 10 days; therefore, it is more likely that varietal differences are from more or less favorable environmental conditions than from relative maturity. For example, in our research at V4 total biomass accumulation was greatest for Terral 48A76, Terral 4857X, Asgrow 46X6, and USDA 25-1. At R2, USDA 25-1 and Terral 4857X, and U of A 5014 produced the greatest soybean biomass. However, planting date and variety influenced biomass accumulation at R5, which was plausibly due to genetic and environmental factors (Table 5). Baker et al. (1989) suggested conducive environmental conditions increase soybean growth, development, and ultimately yield; therefore, biomass accumulation plausibly is dependent on varieties response to the environment. However, biomass accumulation at V4 and R2 did not influence differences in HI or soybean grain yield. Averaged across 4 site years, these data suggest early season biomass accumulation resulted from genetic differences inherent of varieties within slightly different maturity groups. Varieties among differing maturity groups are expected to have morphological differences which plausibly have little effect on yield or HI. According to Cober and Morrison (2010), indeterminate growth habits such as plant height and other agronomic characters are influenced by isogenic lines.

By the time the plants reached mid R5 to R8, differences were observed in biomass accumulation across varieties. Differences in biomass accumulation were variable across plant components and contributed little to soybean yield variability among varieties or planting date (data not shown). Biomass accumulation by soybean components was combined in this study to represent total biomass for each developmental stage from V4 to R8. For the purposes of this study, the focus was to evaluate total biomass accumulation at five developmental stages rather than differences between plant components within each development stage.

Yield components are morphological characteristics of the plant which critically influence yield (Egli, 1998). Soybean yield was analyzed during this experiment to determine the relationship between biomass accumulation, in this case seed, and grain HI. Soybean grain yield was influenced by the main effect of variety. Soybean grain yield for Asgrow 4632 and Terral 48A76 was greater than U of A 5014, Credenz 4748, Terral 4857X, Hornbeck 4953, and USDA 25-1 (Table 6). Harvest index was also influenced by variety (Table 6). Asgrow 4632 and Terral 48A76 produced the greatest HI ranging from 37.2% to 38.2%, which was reflected in the greatest yielding varieties. Asgrow 46X6 produced a similar yield to Terral 48A76 and Asgrow 4632; however, produced a lesser HI at 34.1% indicating a tighter relationship among harvest index to grain yield. These data agree with findings of Bender et al. (2015), suggesting improvements in dry weight HI reflects a shift to modern germplasms in varieties and agronomic practices, increasing biomass production relative to grain yield.

However, this study was confined to the 2017 and 2018, focusing on the general vicinity of Delta Region in Mississippi. The temporal limitation to these 2 years means that findings may not fully capture variation in environmental conditions or agricultural practices that could occur over a longer period and other regions with different soil types, or farming practices.

5 | CONCLUSION

This study evaluated eight soybean varieties at two planting dates in terms of biomass accumulation (at V4, R2, R5, R6, and R8 growth stages), HI, and yield. Yield, HI, and biomass accumulation were unresponsive to changes in planting date but interaction between planting date and variety altered biomass accumulation at mid R5 stage. Variety effect was noted to be significant in affecting yield, HI, and biomass accumulation (at all growth stages except mid R6). Both Terral 48A76 and Asgrow 4632 produced the greatest HI and seed yield. It is also important to consider a variety capable of lesser biomass production and similar grain yield to the aforementioned varieties would produce a greater HI. By mid R6 to R8, differences emerged later in the growing season for varieties capable of greater reproductive biomass production. However, the focus of this research was to determine if biomass production and HI was different for varieties, translating into greater yield potential. Soybean varieties producing greater yields did produce lesser non-reproductive biomass, while producing greater reproductive biomass which was reflected in greater HI values. Differences in grain HI are reflected in the volume of grain produced by some varieties. Varieties producing the greatest yield had HI ratios ranging from 37% to 38%. Greater HI ratios indicate a varieties efficiency in directing biomass production into grain yield. Therefore, soybean yield for the varieties used in this study are influenced by HI rather than overall biomass accumulation. Therefore, variety selection decisions for greater varietal yield potential should be based on HI rather than overall biomass accumulation.

AUTHOR CONTRIBUTIONS

Brian K. Pieralisi: Data curation; formal analysis; investigation; methodology; writing—original draft. **Ramandeep Kumar Sharma**: Writing—original draft; writing—review and editing. **Bobby Golden**: Writing—review and editing. **Jason A. Bond**: Writing—review and editing. **Donald Cook**: Writing—review and editing. **Trent Irby**: Writing—review and editing. **Michael Cox**: Writing—review and editing. **Jagmandeep Dhillon**: Writing—review and editing.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

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